

Ariel Szekely

Contact Information

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Education

- 2020-present **Ph.D. in Computer Science**, *Massachusetts Institute of Technology*, Cambridge, MA
- 2020-2022 **M.S. in Computer Science**, *Massachusetts Institute of Technology*, Cambridge, MA,
Thesis: *σ OS: Elastic Realms for Multi-Tenant Cloud Computing*
GPA: 5.0/5.0
- 2016-2020 **B.S. in Computer Science (Honors)**, *The University of Texas at Austin*, Austin, TX,
Thesis: *Exploring GPU Timing Side-Channels*
GPA: 3.94/4.0

Interests

Distributed Systems, Operating Systems, Security, Concurrency.

Professional Experience

- 2023 **Student Researcher**, *Google Systems Research Group*, Seattle, WA
Designed and evaluated a system for TPU disaggregation for Ads model serving.
- 2019 **Research Assistant**, *The University of Texas at Austin*, Austin, TX
Developed the Telekine GPU timing side-channel attack for our NSDI '20 paper.
- 2018 **Software Engineer Intern**, *Bloomberg L.P.*, New York City, NY
Back-end development.
- 2017 **Software Engineer Intern**, *AT&T*, Dallas, TX
Full-stack development.

Publications

- Eurosys '27 (under submission) **Fast end-to-end cloud application cold-start with co-sandboxes**, Ariel Szekely, Hannah Gross, Robert Morris, Frans Kaashoek
Designed and implemented co-sandboxes, a new scriptable interface for cloud developers to specify their application's initialization routine to the provider in order to overlap application initialization with infrastructure setup costs. Co-sandboxes reduce cold-start latency of serverless applications by 1.6-1.8X and microservices like etcd and memcached by 1.7X.
- SOSP '24 **Unifying serverless and microservice workloads with SigmaOS**, Ariel Szekely, Adam Belay, Robert Morris, M. Frans Kaashoek
Designed and implemented SigmaOS, a novel cloud operating system and cluster manager which supports both microservices and serverless applications. SigmaOS's API enables applications to warm-start in under 2ms and cold-start in under 8ms. [GitHub]

- ASPLOS '23 **Towards a machine learning-assisted kernel with LAKE**, *Henrique Fingler, Isha Tarte, Hangchen Yu, Ariel Szekely, Bodun Hu, Aditya Akella, Christopher J. Rossbach*
Rewrote the Linux Kernel Samepage Merging (KSM) module to benefit from hardware acceleration.
- NSDI '20 **Telekine: Secure computing with cloud GPUs**, *Tyler Hunt, Zhipeng Jia, Vance Miller, Ariel Szekely, Yige Hu, Christopher J. Rossbach, Emmett Witchel*
Built a timing side-channel attack against a GPU Trusted Execution Environment (TEE), allowing the attacker to learn the class of images passed through ResNet by timing the GPU kernels.

Awards

- 2022 NSF Graduate Research Fellowship
- 2020 MIT Lemelson Presidential Fellowship
- 2020 MIT EECS Departmental Fellowship
- 2020 Dean's Honored Graduate, UT Austin College of Natural Sciences
- 2020 Research Distinction, UT Austin College of Natural Sciences
- 2020 NSF Graduate Research Fellowship Honorable Mention
- 2020 CRA Outstanding Undergraduate Researcher Honorable Mention
- 2019 NSF REU
- 2019 UT Endowed Presidential Scholarship
- 2018,19,20 UT College of Natural Sciences College Scholar
- 2018 Louis E. Rosier Memorial Scholarship
- 2016 UT CNS Freshman Scholarship
- 2016 National Merit Scholarship

Teaching Experience

- 2023 **TA: Computer Systems Security**, *Massachusetts Institute of Technology*
Instructors: Nickolai Zeldovich
- 2022 **TA: Operating Systems Engineering**, *Massachusetts Institute of Technology*
Instructors: Frans Kaashoek, Robert Morris
- 2020 **TA: Concurrency (Honors)**, *The University of Texas at Austin*
Instructor: Christopher J. Rossbach
- 2020 **TA: Intro to CS Research (Honors)**, *The University of Texas at Austin*
Instructor: Calvin Lin
- 2019 **TA: Intro to CS Research (Honors)**, *The University of Texas at Austin*
Instructor: Calvin Lin

Service

- 2022-present **EECS Communication Lab Fellow**, *Massachusetts Institute of Technology, Cambridge, MA*
- 2021 **SOSP '21 Artifact Evaluation Committee**, *SOSP 2021, Kolbenz, Germany*
- 2020-present **MIT Graduate Application Assistance Program Mentor**, *Massachusetts Institute of Technology, Cambridge, MA*

Languages

English Fluent

Spanish Fluent
French Intermediate